

To prove: $\widehat{AXB} = 90^\circ$.

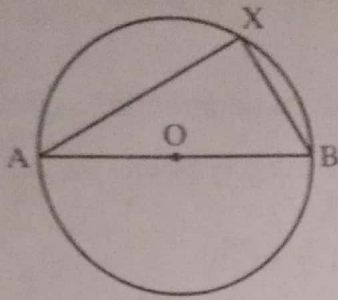


Fig. 2.18

Proof:

With the lettering in Fig. 2.18,

$$\widehat{AOB} = 2\widehat{AXB} \quad (\text{angle at centre} = 2 \times \text{angle at circumference})$$

$$\text{but } \widehat{AOB} = 180^\circ \quad (\text{straight angle})$$

$$\therefore 2\widehat{AXB} = 180^\circ$$

$$\therefore \widehat{AXB} = 90^\circ$$

Example 4

In Fig. 2.19, $\overline{PQ}$ is a diameter of circle PMQN, centre O. If $\widehat{PQM} = 63^\circ$, find $\widehat{QNM}$.

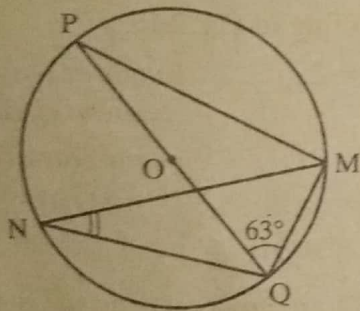


Fig. 2.19

In $\triangle QPM$,

$$\widehat{PMQ} = 90^\circ \quad (\text{angle in semicircle})$$

$$\therefore \widehat{QPM} = 180^\circ - 90^\circ - 63^\circ$$

$$= 27^\circ \quad (\text{angle sum of } \triangle)$$

$$\therefore \widehat{QNM} = 27^\circ \quad (\text{in same segment as } \widehat{QPM})$$

Exercise 2c

Find the lettered angles in each of the following. Where a point O is given it is the centre of the circle.

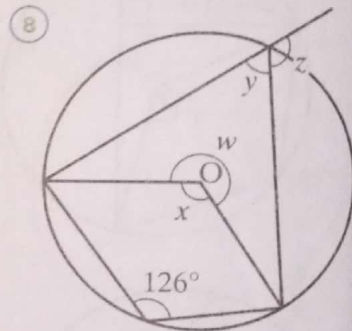
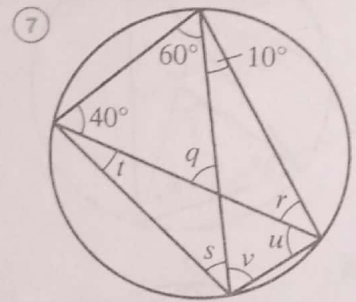
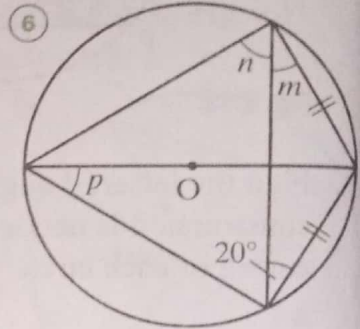
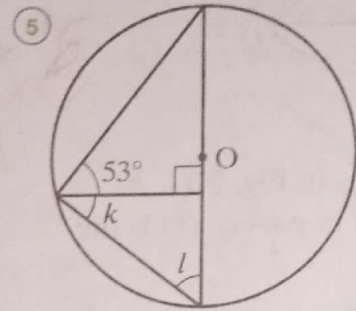
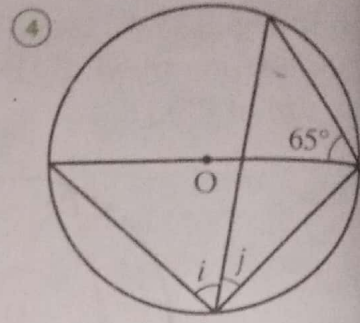
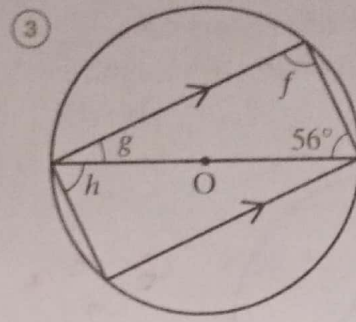
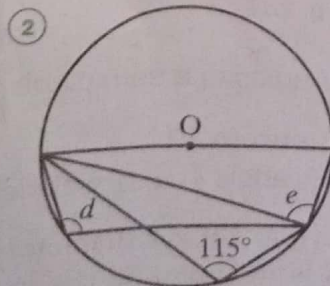
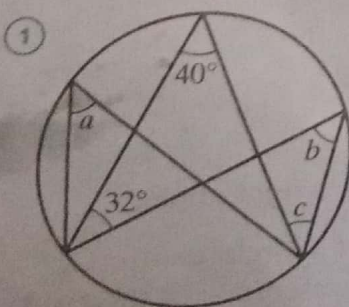


Fig. 2.20

2.3 Cyclic quadrilaterals

The vertices of a **cyclic quadrilateral** lie on the circumference of a circle. In Fig. 2.21, ABCD and PQRS are cyclic quadrilaterals.

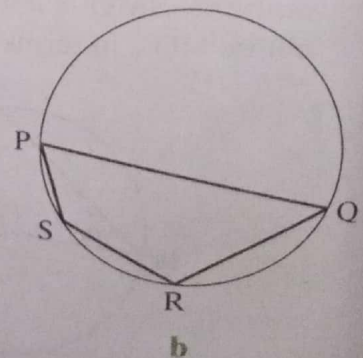
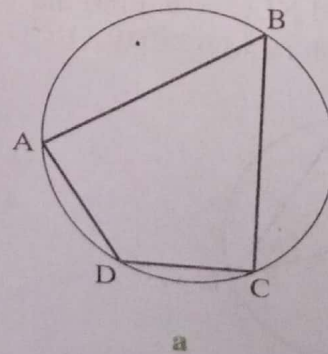


Fig. 2.21

Notice that opposite angles of a cyclic quadrilateral lie in opposite segments of a circle (Fig. 2.22).

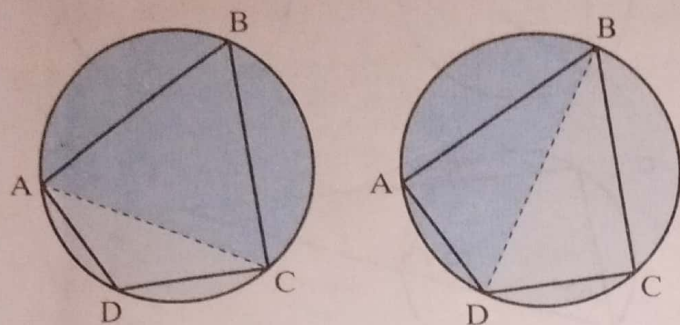


Fig. 2.22

Theorem 14

The opposite angles of a cyclic quadrilateral are supplementary.

or

Angles in opposite segments are supplementary.

Note: the sum of **supplementary angles** is 180° .

Given: a cyclic quadrilateral ABCD.

To prove: $\widehat{BAD} + \widehat{BCD} = 180^\circ$.

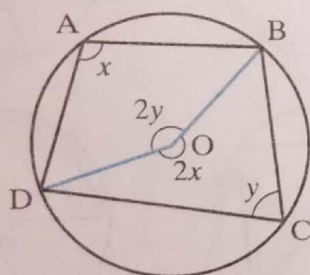


Fig. 2.23

Construction: Join B and D to the centre O of circle ABCD.

Proof:

With the lettering in Fig. 2.23,

$$\widehat{BOD} = 2y \quad (\text{angle at centre} = 2 \times \text{angle at circumference})$$

$$\text{reflex } \widehat{BOD} = 2x \quad (\text{same reason})$$

$$\therefore 2x + 2y = 360^\circ \quad (\text{angles at a point})$$

$$\therefore x + y = 180^\circ$$

$$\therefore \widehat{BAD} + \widehat{BCD} = 180^\circ$$

It follows from Theorem 14 that **the exterior angle of a cyclic quadrilateral is equal to the interior opposite angle.**

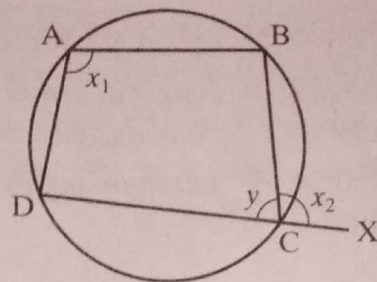


Fig. 2.24

With the lettering in Fig. 2.24,

$$x_1 + y = 180^\circ \quad (\text{opp. angles of cycl. quad.})$$

$$x_2 + y = 180^\circ \quad (\text{straight angle})$$

$$\therefore x_1 = x_2 \quad (= 180^\circ - y)$$

$$\therefore \widehat{BCX} = \widehat{BAD}$$

Example 5

In Fig. 2.25, $\overline{CE}$ is a diameter of circle ABCDE. If $\widehat{ABC} = 127^\circ$, find $\widehat{ACE}$.

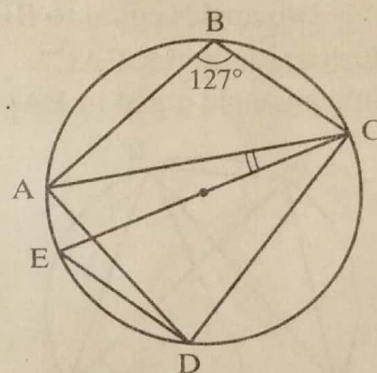


Fig. 2.25

$$\widehat{ADC} = 180^\circ - 127^\circ \quad (\text{opp. angles of cycl. quad.})$$

$$= 53^\circ$$

$$\widehat{EDC} = 90^\circ \quad (\text{angle in semicircle})$$

$$\therefore \widehat{EDA} = 90^\circ - 53^\circ$$

$$= 37^\circ$$

$$\therefore \widehat{ACE} = 37^\circ \quad (\text{in same segment as } \widehat{EDA})$$

Example 6

In Fig. 2.26, P, Q, R, S are points on a circle centre O. $\overline{QP}$ is produced to X. If $\widehat{XPS} = 77^\circ$ and $\widehat{PSO} = 68^\circ$, find $\widehat{PQO}$.

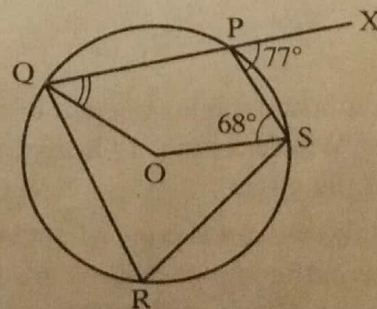


Fig. 2.26

$$\begin{aligned}\widehat{QRS} &= 77^\circ \quad (= \text{ext. angle of cycl. quad.}) \\ \therefore \widehat{QOS} &= 2 \times 77^\circ \quad (\text{angle at centre} = 2 \times \text{angle at circumference}) \\ &= 154^\circ \\ \widehat{QPS} &= 180^\circ - 77^\circ \quad (\text{straight angle}) \\ &= 103^\circ\end{aligned}$$

In quad. PQOS,

$$\widehat{PQO} = 360^\circ - 154^\circ - 103^\circ - 68^\circ \quad (\text{angle sum of quad.})$$

$$\therefore \widehat{PQO} = 35^\circ$$

Exercise 2d

① (oral) Use Fig. 2.27 to answer the following.

- i Which arc subtends $\widehat{ABE}$?
ii Name an angle equal to $\widehat{ABE}$.
- i Which arc subtends $\widehat{BEC}$?
ii Name two angles equal to $\widehat{BEC}$.
- i Which arc subtends $\widehat{EAC}$?
ii Name an angle equal to $\widehat{EAC}$.

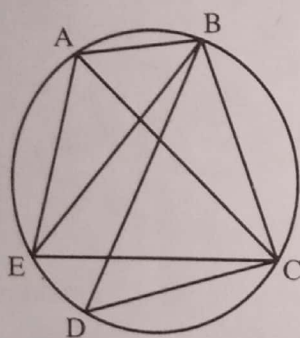


Fig. 2.27

② In Fig. 2.28, $\overline{AB}$ is a diameter of semicircle ABCD. If $\widehat{ABD} = 16^\circ$, calculate $\widehat{BCD}$. (Hint: Join $\overline{CA}$).

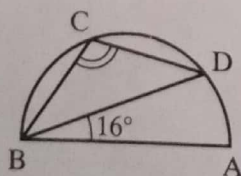


Fig. 2.28

③ Find the marked angles in each of the following. Where a point O is given it is the centre of the circle.

(Hint: Make a careful copy of each figure and write in the angles as they are found. Some constructions may be necessary.)

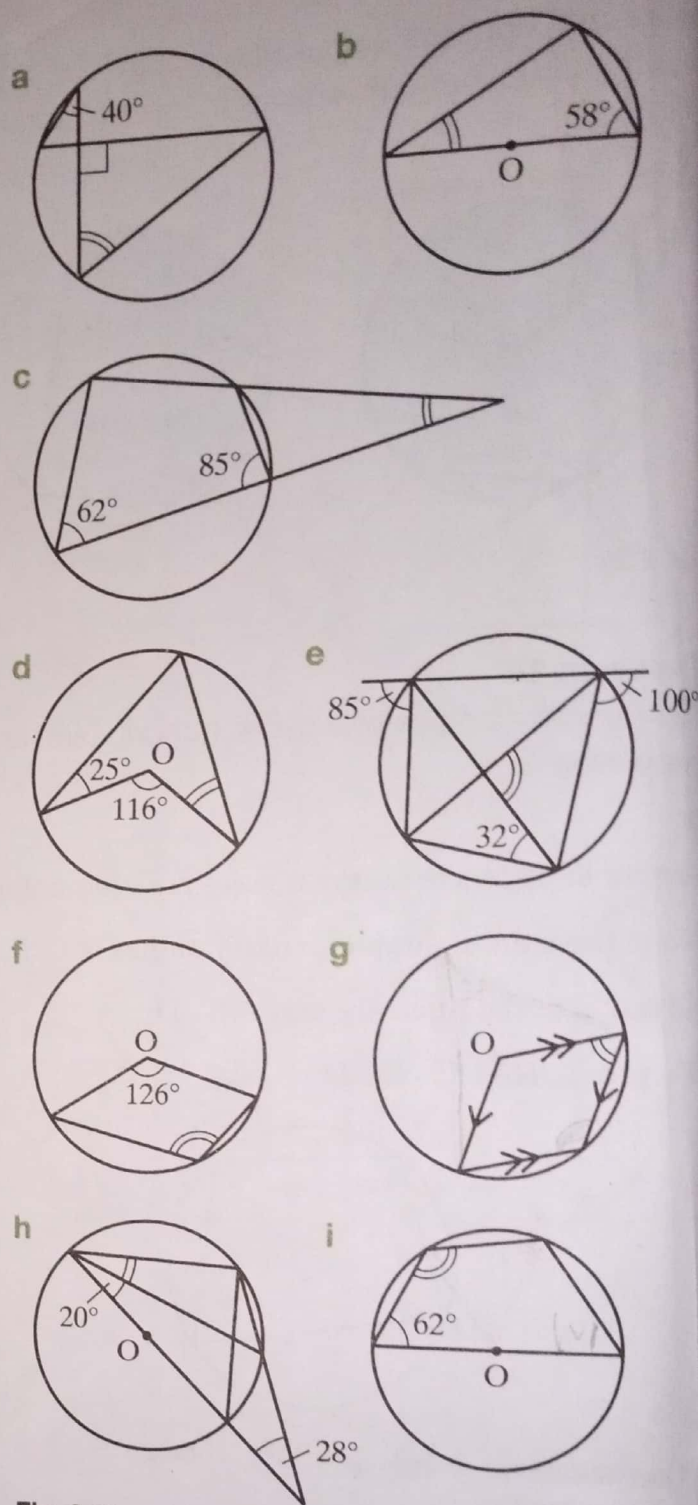


Fig. 2.29

④ In Fig. 2.30, A, B, C, D are points on a circle such that $\widehat{ABC} = 102^\circ$. $\overline{CD}$ is produced to E so that $\widehat{AED} = 47^\circ$. Calculate $\widehat{EAD}$.

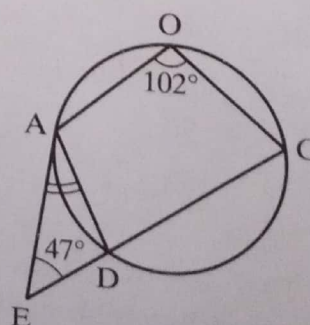


Fig. 2.30

- 5 In Fig. 2.31, ABCD is a cyclic quadrilateral such that $\overline{AB} \parallel \overline{DC}$. If $\widehat{ACD} = 35^\circ$ and $\widehat{DBC} = 72^\circ$, calculate the sizes of the angles of the trapezium.

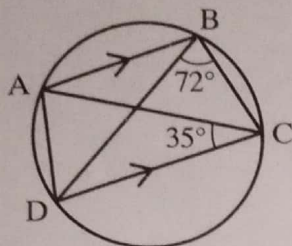


Fig. 2.31

- 6 In Fig. 2.32, $\widehat{PQR} = 80^\circ$ and $\widehat{SRT} = 20^\circ$. What size is $\widehat{PXR}$? [WAEC]

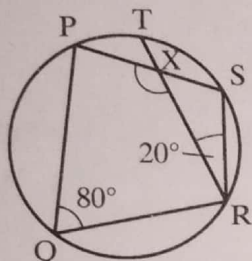


Fig. 2.32

- 7 In Fig. 2.33, O is the centre of circle MLY. If $\widehat{OLY} = 50^\circ$ and $\widehat{OMY} = 15^\circ$, calculate $\widehat{MOL}$. [WAEC]

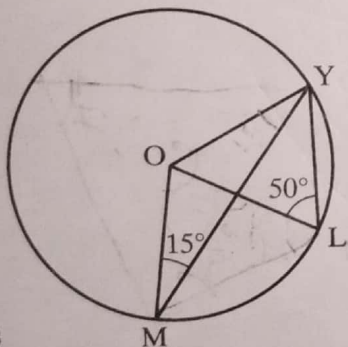


Fig. 2.33

- 8 In Fig. 2.34, PQRS is a cyclic quadrilateral such that $\widehat{QPR} = 18^\circ$ and $\widehat{RQS} = 42^\circ$. If $\widehat{PSR} = 78^\circ$, calculate the angles of $\triangle PQS$. What kind of triangle is $\triangle PQS$?

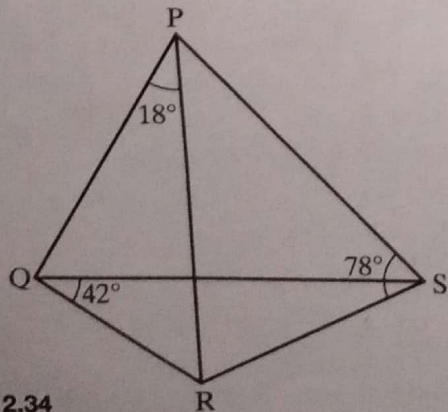


Fig. 2.34

- 9 a Prove that the opposite angles of a cyclic quadrilateral are supplementary.
b In Fig. 2.35, calculate the value of x giving a reason for each step in your answer. [WAEC]

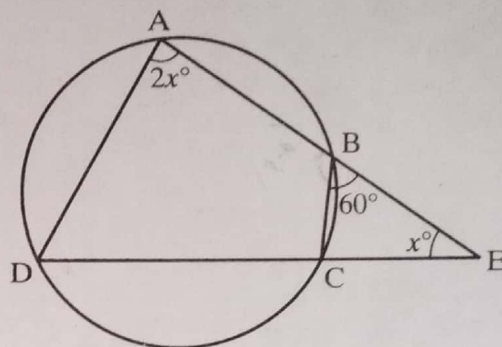


Fig. 2.35

- 10 a In Fig. 2.36, O is the centre of the circle and A, B and P are points on the circumference. Prove that $\widehat{AOB} = 2\widehat{APB}$.

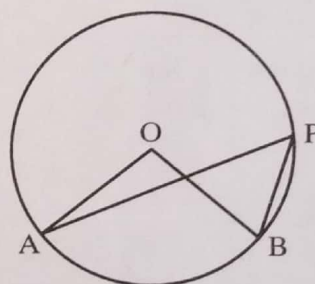


Fig. 2.36

- b In Fig. 2.37, P, Q, R, S are points on a circle such that $\widehat{PQS} = \widehat{PRQ}$.

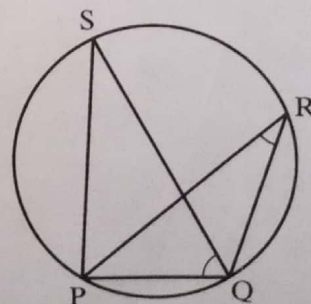


Fig. 2.37

- i Prove that $|PS| = |PQ|$.
ii Hence, if SQ is a diameter of the circle and $\overline{QR}$ is produced to Z, determine $\widehat{SQP}$ and $\widehat{SRZ}$, giving reasons.

[WAEC]

SUMMARY

Remember the following theorems when calculating angles between lines that are drawn inside circles.

Theorem	Visual aid	Notes
A straight line from the centre of a circle that bisects a chord, is at right angles to the chord.		$\widehat{OMA} = \widehat{OMB} = 90^\circ$ (The chord must not be a diameter.) See Examples 1 and 2.
The angle subtended at the centre of a circle is twice that subtended at the circumference.		$\widehat{AOB} = 2\widehat{APB}$ Reflex $\widehat{AOB} = 2\widehat{AQB}$ (Fig. 2.9 shows other illustrations of the same theorem.) See Example 3.
Angles in the same segment are equal.		$\widehat{APB} = \widehat{AQB}$ Similarly, $\widehat{PAQ} = \widehat{PBQ}$ (Note that a chord cuts a circle into a major segment and a minor segment as shown in Fig. 2.17.)
The angle in a semicircle is a right angle.		$\widehat{AXB} = 90^\circ$ (This is a special case of the theorem 2 above.) See Example 4.
Opposite angles of a cyclic quadrilateral are supplementary (i.e. add up to 180°).		$\widehat{BAD} + \widehat{BCD} = 180^\circ$ See Examples 5 and 6.

3

Algebraic processes (1) Quadratic equations

OBJECTIVES

By the end of this chapter you should be able to:

- 1 Solve quadratic equations by
 - factorisation
 - using perfect squares
 - completing the square
 - using the quadratic formula
- 2 Form a quadratic equation given the sum and the product of its roots
- 3 Solve word problems by forming and solving suitable quadratic equations.

Teaching and learning materials

Teacher: Poster showing the quadratic formula.

3.1 Quadratic equations (revision)

If the product of two real numbers is 0, then one of the numbers (or both of them) must be 0.
In general,

$$\text{if } a \times b = 0 \\ \text{either } a = 0 \quad \text{or} \quad b = 0$$

Example 1

Solve the equation $(x - 4)(x + 9) = 0$.

$$\begin{aligned} \text{If } (x - 4)(x + 9) &= 0 \\ \text{then either } x - 4 &= 0 \quad \text{or} \quad x + 9 = 0 \\ x &= 4 \quad \text{or} \quad x = -9 \end{aligned}$$

Example 2

Solve $2x^2 + 13x = 15$.

$$\begin{aligned} \text{If } 2x^2 + 13x &= 15 \\ \Rightarrow 2x^2 + 13x - 15 &= 0 \\ \Rightarrow (2x + 15)(x - 1) &= 0^* \\ \Rightarrow 2x + 15 = 0 \quad \text{or} \quad x - 1 &= 0 \\ \Rightarrow 2x = -15 \quad \text{or} \quad x &= 1 \\ \Rightarrow x = -7\frac{1}{2} \quad \text{or} \quad x &= 1 \end{aligned}$$

***Note:** Factorisation of quadratics is fully explained in Chapter 10 of *New General Mathematics* for Senior Secondary Schools Book 1.

Exercise 3a (Revision)

Solve the following.

- ① $(a + 5)(a - 6) = 0$
- ② $(2x - 9)(x + 8) = 0$
- ③ $(5 - b)(12 - 5b) = 0$
- ④ $(2y + 7)^2 = 0$
- ⑤ $c^2 + 10c = 0$
- ⑥ $6z = 5z^2$
- ⑦ $d^2 - 5d + 6 = 0$
- ⑧ $p^2 - 10p - 24 = 0$
- ⑨ $15 - 2e - e^2 = 0$
- ⑩ $q^2 - 11q + 24 = 0$
- ⑪ $f^2 + 25f + 150 = 0$
- ⑫ $r^2 + 25r - 150 = 0$
- ⑬ $g^2 + 4g = 21$
- ⑭ $s^2 = 2s + 15$
- ⑮ $2h^2 - 5h + 3 = 0$
- ⑯ $2t^2 - t - 1 = 0$
- ⑰ $3k^2 + 2k - 1 = 0$
- ⑱ $35 + 18u - 5u^2 = 0$
- ⑲ $10m^2 = 41m + 45$
- ⑳ $21v = 8v^2 - 9$

3.2 Equations with irrational roots

If $m^2 = 16$, then $m = \pm 4$.

Hence, if $(x - 3)^2 = 16$

then $x - 3 = \pm 4$

and $x = 3 \pm 4$

$$\begin{aligned} &= 7 \text{ or } -1 \quad (3 + 4 = 7 \\ &\quad 3 - 4 = -1) \end{aligned}$$

Example 3

Solve the equation $(x + 3)^2 = 7$.

$$\begin{aligned}\text{If } (x + 3)^2 &= 7 \\ \text{then } x + 3 &= \pm\sqrt{7} \\ x &= -3 \pm \sqrt{7}\end{aligned}$$

Since $\sqrt{7}$ is an irrational number, the roots of the equation $(x + 3)^2 = 7$ are irrational. The roots may be found approximately by writing $\pm\sqrt{7}$ as ± 2.65 .

Exercise 3b

Solve the following equations. If an equation has irrational roots, leave the answer in the form given in Example 3.

- | | |
|--|---|
| ① $(x - 2)^2 = 9$ | ② $(x - 7)^2 = 4$ |
| ③ $(x + 3)^2 = 4$ | ④ $(x + 2)^2 = 25$ |
| ⑤ $(x - 1)^2 = 2$ | ⑥ $(x + 4)^2 = 3$ |
| ⑦ $(x - 3)^2 = 5$ | ⑧ $(x + 2)^2 = 2$ |
| ⑨ $(x - 2)^2 = \frac{1}{4}$ | ⑩ $(x - 6)^2 = 36$ |
| ⑪ $(x - 4)^2 = 10$ | ⑫ $(x + 5)^2 = \frac{1}{9}$ |
| ⑬ $(x + 3)^2 = 49$ | ⑭ $(x - 1)^2 = 7$ |
| ⑮ $(x - 8)^2 = 3$ | ⑯ $(x - 1)^2 = \frac{9}{25}$ |
| ⑰ $(x + 1)^2 = 2\frac{1}{4}$ | ⑲ $(x + 7)^2 = 6$ |
| ⑱ $(x + \frac{1}{3})^2 = \frac{4}{9}$ | ⑳ $(x + 9)^2 = 3$ |
| ㉑ $(x - 6)^2 = 5$ | ㉒ $(x - 2\frac{1}{2})^2 = 6\frac{1}{4}$ |
| ㉓ $(x + 10)^2 = 8$ | ㉔ $(x - 6)^2 = 2\frac{7}{9}$ |
| ㉕ $(x + 1\frac{1}{4})^2 = 1\frac{9}{16}$ | |

3.3 Completing the square

Example 4

What must be added to $x^2 + 6x$ to make the expression a perfect square?

Suppose $x^2 + 6x + k$ is a perfect square and that it is equal to $(x + a)^2$.

$$\begin{aligned}\text{i.e. Let } x^2 + 6x + k &= (x + a)^2 \\ \text{then } x^2 + 6x + k &= x^2 + 2ax + a^2\end{aligned}$$

By comparing coefficients of x ,

$$2a = 6$$

$$a = 3$$

By comparing the constant terms,

$$k = a^2$$

$$k = 3^2 = 9$$

Therefore 9 must be added to the expression.

$$\text{Check: } x^2 + 6x + 9 = (x + 3)^2$$

In practice, the quantity to be added is **the square of half of the coefficient of x** (or whatever letter is involved). In Example 4, the coefficient of x is 6, half of 6 is 3 and the square of 3 is 9. Hence 9 must be added.

Example 5

What must be added to $d^2 - 5d$ to make the expression a perfect square?

Factorise the result.

The coefficient of d is -5 . Half of -5 is $-\frac{5}{2}$.

$$\left(-\frac{5}{2}\right)^2 = +\frac{25}{4}$$

$\frac{25}{4}$ must be added.

$d^2 - 5d + \frac{25}{4}$ is a perfect square.

$$\begin{aligned}d^2 - 5d + \frac{25}{4} &= \left(d - \frac{5}{2}\right)^2 \\ &= \left(d - 2\frac{1}{2}\right)^2\end{aligned}$$

Example 6

Add a term to $n^2 + 1\frac{1}{2}n$ to make the expression a perfect square. Express the result as the square of a bracketed expression.

The coefficient of n is $+1\frac{1}{2}$.

Half of $+1\frac{1}{2} = \frac{1}{2} \times \frac{3}{2} = \frac{3}{4}$ and $\left(\frac{3}{4}\right)^2 = \frac{9}{16}$.

$\frac{9}{16}$ must be added.

$$n^2 + 1\frac{1}{2}n + \frac{9}{16} = \left(n + \frac{3}{4}\right)^2$$

Examples 5 and 6 can be checked by squaring out the bracket in the final result.

Exercise 3c

In each of the following, add the term that makes the given expression into a perfect square. Write the result as the square of a bracketed expression.

- | | |
|-------------------------|-------------------------|
| ① $a^2 + 8a$ | ② $b^2 + 10b$ |
| ③ $c^2 - 4c$ | ④ $d^2 - 6d$ |
| ⑤ $x^2 + 5x$ | ⑥ $y^2 - 3y$ |
| ⑦ $z^2 - 7z$ | ⑧ $m^2 + 2m$ |
| ⑨ $n^2 - n$ | ⑩ $u^2 - \frac{1}{2}u$ |
| ⑪ $v^2 + \frac{1}{4}v$ | ⑫ $h^2 + \frac{2}{3}h$ |
| ⑬ $k^2 - 1\frac{1}{3}k$ | ⑭ $g^2 - 4\frac{2}{3}g$ |
| ⑮ $a^2 + \frac{3}{5}a$ | ⑯ $b^2 - \frac{4}{5}b$ |
| ⑰ $c^2 - 1\frac{1}{2}c$ | ⑱ $m^2 - 8m$ |
| ⑲ $m^2 - 8mn$ | ⑳ $a^2 - 6ad$ |
| ㉑ $x^2 + 10xy$ | ㉒ $m^2 + 3mn$ |
| ㉓ $u^2 - 1\frac{3}{5}u$ | ㉔ $v^2 - \frac{3}{4}v$ |

Example 7

Solve the equation $x^2 - 8x + 3 = 0$.

The LHS (Left Hand Side) does not factorise. So rearrange the equation to make the LHS a perfect square.

$$x^2 - 8x + 3 = 0$$

Subtract 3 from both sides.

$$x^2 - 8x = -3$$

Add 16 to both sides.

$$x^2 - 8x + 16 = -3 + 16$$

$$\Rightarrow (x - 4)^2 = 13$$

$$\Rightarrow x - 4 = \pm\sqrt{13}$$

$$\Rightarrow x = 4 \pm\sqrt{13}$$

Example 8

Solve the equation $a^2 + 3a - 2 = 0$.

The LHS does not factorise.

$$a^2 + 3a - 2 = 0$$

$$\Rightarrow a^2 + 3a = 2$$

Add the square of $\frac{3}{2}$ to both sides.

$$a^2 + 3a + \left(\frac{3}{2}\right)^2 = 2 + \frac{9}{4}$$

$$= \frac{8 + 9}{4}$$

$$\Rightarrow \left(a + \frac{3}{2}\right)^2 = \frac{17}{4}$$

$$\Rightarrow a + \frac{3}{2} = \pm\sqrt{\frac{17}{4}}$$

$$= \pm\frac{\sqrt{17}}{2}$$

$$\Rightarrow a = -\frac{3}{2} \pm \frac{\sqrt{17}}{2}$$

$$= \frac{-3 \pm \sqrt{17}}{2}$$

If the LHS of the equation factorises, use the method of factorisation rather than completing the square. For example,

$$x^2 - 7x + 10 = 0$$

$$\Rightarrow (x - 5)(x - 2) = 0$$

$$\Rightarrow x = 5 \text{ or } 2$$

rather than

$$x^2 - 7x + 10 = 0$$

$$\Rightarrow x^2 - 7x = -10$$

$$\Rightarrow x^2 - 7x + \left(\frac{7}{2}\right)^2 = -10 + \frac{49}{4}$$

$$= \frac{-40 + 49}{4}$$

$$\Rightarrow \left(x - \frac{7}{2}\right)^2 = \frac{9}{4}$$

$$\Rightarrow x - \frac{7}{2} = \pm\frac{3}{2}$$

$$\Rightarrow x = \frac{7}{2} \pm \frac{3}{2}$$

$$= \frac{10}{2} \text{ or } \frac{4}{2}$$

$$= 5 \text{ or } 2$$

Exercise 3d

Solve the following equations. Factorise where possible. Otherwise, solve by completing the square, leaving the answers in the form of Examples 7 and 8.

- | | |
|-----------------------|------------------------|
| ① $a^2 + 4a - 21 = 0$ | ② $b^2 - b - 12 = 0$ |
| ③ $c^2 - 4c - 2 = 0$ | ④ $d^2 + 2d - 2 = 0$ |
| ⑤ $n^2 + 4n + 4 = 0$ | ⑥ $p^2 - 10p + 15 = 0$ |

- ⑦ $q^2 + 10q + 22 = 0$ ⑧ $t^2 - 6t + 9 = 0$
 ⑨ $m^2 + 6m + 7 = 0$ ⑩ $y^2 - 3y + 1 = 0$
 ⑪ $z^2 - 5z + 6 = 0$ ⑫ $h^2 + 5h + 4 = 0$
 ⑬ $k^2 - 5k + 2 = 0$ ⑭ $g^2 + 5g + 2 = 0$
 ⑮ $x^2 - 8x - 1 = 0$ ⑯ $a^2 - a - 1 = 0$
 ⑰ $b^2 + b - 3 = 0$ ⑱ $y^2 + 7y - 30 = 0$
 ⑲ $m^2 - 7m + 11 = 0$ ⑳ $x^2 + 3x - 2 = 0$
 ㉑ $x^2 - 10x + 25 = 0$ ㉒ $v^2 + 9v + 19 = 0$
 ㉓ $n^2 - 12n + 1 = 0$ ㉔ $u^2 - 14u - 3 = 0$

3.4 The quadratic formula

The **general form of a quadratic equation** is $ax^2 + bx + c = 0$. The roots of the general quadratic equation are found by completing the square.

$$\begin{aligned}
 & ax^2 + bx + c = 0 \\
 \Rightarrow & x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \\
 \Rightarrow & x^2 + \frac{b}{a}x = -\frac{c}{a} \\
 \Rightarrow & x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2 \\
 & \quad = -\frac{c}{a} + \frac{b^2}{4a^2} \\
 \Rightarrow & \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} \\
 \Rightarrow & x + \frac{b}{2a} = \pm \sqrt{\frac{b^2 - 4ac}{4a^2}} \\
 & \quad = \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\
 \Rightarrow & x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\
 \Rightarrow & x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
 \end{aligned}$$

This is called the **quadratic formula**. Use the formula to solve *any* quadratic equation.

Example 9

Find, correct to 2 decimal places, the roots of the equation $3x^2 - 5x - 7 = 0$.

Comparing $3x^2 - 5x - 7 = 0$ with

$$ax^2 + bx + c = 0;$$

$$a = 3, b = -5, c = -7.$$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4 \times 3 \times (-7)}}{2 \times 3}$$

$$= \frac{5 \pm \sqrt{25 + 84}}{6}$$

$$= \frac{5 \pm \sqrt{109}}{6}$$

$$= \frac{5 \pm 10.44}{6}$$

$$= \frac{15.44}{6} \quad \text{or} \quad \frac{-5.44}{6}$$

$$\approx 2.57 \quad \text{or} \quad -0.91$$

Exercise 3e

Use the quadratic formula to solve the following equations. Give the roots correct to 2 decimal places where necessary. Use factorisation to check the results of the first ten.

- ① $x^2 + 5x + 6 = 0$ ② $x^2 - 5x + 4 = 0$
 ③ $x^2 - 4x - 5 = 0$ ④ $2x^2 + 5x + 3 = 0$
 ⑤ $3x^2 - 4x + 1 = 0$ ⑥ $3x^2 - 5x - 2 = 0$
 ⑦ $5x^2 - 3x - 2 = 0$ ⑧ $4x^2 + 7x - 2 = 0$
 ⑨ $6x^2 + 13x + 6 = 0$ ⑩ $3x^2 - 13x - 10 = 0$
 ⑪ $x^2 + 3x + 1 = 0$ ⑫ $x^2 - 2x - 4 = 0$
 ⑬ $2x^2 + 7x - 3 = 0$ ⑭ $3x^2 - 5x - 3 = 0$
 ⑮ $5x^2 - 6x - 3 = 0$ ⑯ $5x^2 + 8x - 2 = 0$
 ⑰ $3x^2 + 7x + 3 = 0$ ⑱ $3x^2 - 12x + 10 = 0$
 ⑲ $3x^2 - 8x + 2 = 0$ ㉑ $5x^2 + 3x - 3 = 0$

3.5 Sum and product of roots

As we have seen earlier, the **roots** of an equation are the values of the unknown that make the equation true. For example, +9 and -1 are the roots of the equation $x^2 - 8x - 9 = 0$.

In this section we show that it is possible to find information about the roots of an equation without solving it. If α and β are the roots of an equation in x , then:

$$\begin{aligned}(x - \alpha)(x - \beta) &= 0 \\ x^2 - \alpha x - \beta x + \alpha\beta &= 0 \\ x^2 - (\alpha + \beta)x + \alpha\beta &= 0 \quad (1)\end{aligned}$$

In equation (1), $(\alpha + \beta)$ is the **sum of the roots** and $\alpha\beta$ is the **product of the roots**. Examples 10 and 11 show how to apply this result.

Example 10

Find the sum and the product of the roots of the equation

a $5x^2 - 4x - 9 = 0$ **b** $2x^2 + 9x = 6$.

a $5x^2 - 4x - 9 = 0$

Dividing throughout by 5

$$x^2 - \frac{4}{5}x - \frac{9}{5} = 0$$

Comparing this with equation (1)

$$(\alpha + \beta) = \frac{4}{5} \text{ and } \alpha\beta = -\frac{9}{5}$$

The sum of the roots is $\frac{4}{5}$ and their product is $-\frac{9}{5}$.

b $2x^2 + 9x = 6$

$$2x^2 + 9x - 6 = 0$$

Dividing throughout by 2

$$x^2 + \frac{9}{2}x - 3 = 0$$

Comparing this with equation (1)

$$(\alpha + \beta) = -\frac{9}{2} \text{ and } \alpha\beta = -3$$

The sum of the roots is $-\frac{9}{2}$ and their product is -3 .

Example 11

Find the equation whose sum and product of roots are respectively **a** $-5, 6$, **b** $1.5, -1.2$.

Write each answer in the form $ax^2 + bx + c = 0$.

Let the sum and product of roots be $(\alpha + \beta)$ and $\alpha\beta$ as in equation (1) above, then

a $(\alpha + \beta) = -5$ and $\alpha\beta = 6$

Substitute these values into

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

to give:

$$x^2 - (-5)x + 6 = 0$$

$$x^2 + 5x + 6 = 0$$

b $(\alpha + \beta) = 1.5$ and $\alpha\beta = -1.2$

Substitute these values into

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

to give:

$$x^2 - (1.5)x - 1.2 = 0$$

$$x^2 - 1.5x - 1.2 = 0$$

$$x^2 - \frac{3}{2}x - \frac{6}{5} = 0$$

Multiply by 10 throughout:

$$10x^2 - 15x + 12 = 0$$

Exercise 3f

① Use the method of Example 10 to find the sums and products of the roots of the following equations.

a $2x^2 - 10x + 7 = 0$

b $3x^2 + x - 5 = 0$

c $4x^2 = 3x + 1$

d $y^2 + 5y = 1$

e $a(a - 1) = 5$

f $x + \frac{1}{x} = 3$

② Use the method of Example 11 to find equations in the form $ax^2 + bx + c = 0$ whose sums and products of roots are respectively:

a $3, 4$

b $-5, 6$

c $\frac{3}{2}, -\frac{5}{2}$

d $-\frac{7}{3}, 0$

e $0, -7$

f $1.2, 0.8$

3.6 Word problems leading to quadratic equations

Example 12

Find two numbers whose difference is 5 and whose product is 266.

Let the smaller number be x .
Then the larger number is $x + 5$.
Their product is $x(x + 5)$.

$$\begin{aligned}\text{Hence } x(x + 5) &= 266 \\ \Rightarrow x^2 + 5x - 266 &= 0 \\ \Rightarrow (x - 14)(x + 19) &= 0 \\ \Rightarrow x &= 14 \text{ or } -19\end{aligned}$$

The other number is $14 + 5$ or $-19 + 5$
i.e. 19 or -14 .

The two numbers are 14 and 19, or -19 and -14 .

Check: $14 \times 19 = 266$ and $-19 \times -14 = 266$

Example 13

A woman is 4 times older than her child. 5 years ago the product of their ages was 175. Find their present ages.

Let the child's age be x years.
Then the mother's age is $4x$ years.
5 years ago, the child's age was $(x - 5)$ years
and the mother's age was $(4x - 5)$ years.
The product of their ages was $(x - 5)(4x - 5)$.

$$\begin{aligned}\text{Hence } (x - 5)(4x - 5) &= 175 \\ \Rightarrow 4x^2 - 25x + 25 &= 175 \\ \Rightarrow 4x^2 - 25x - 150 &= 0 \\ \Rightarrow (4x + 15)(x - 10) &= 0 \\ \Rightarrow x &= 10 \text{ or } -\frac{15}{4}\end{aligned}$$

However, $-\frac{15}{4}$ is not sensible for an age (ages cannot be negative). Therefore the child is 10 years old and the mother is 40 years old.

Check: $(40 - 5)(10 - 5) = 35 \times 5 = 175$

Notice, in Example 13, that one of the roots does not give a sensible answer. Disregard any root which is not sensible in terms of the given question.

Exercise 3g

- ① Find two numbers which differ by 4 and whose product is 45.
- ② The width of a classroom is 4 m less than the length. Its area is 45 m^2 . Find the dimensions of the classroom.
- ③ Two numbers have a difference of 3. The sum of their squares is 89. Find the numbers.
- ④ Two square rooms have a total floor area of 89 m^2 . One room is 3 m longer each way than the other. Find the dimensions of the two rooms.
- ⑤ A girl is 6 years younger than her brother. The product of their ages is 135. Find their ages.
- ⑥ The ages of two children are 11 and 8 years. In how many years' time will the product of their ages be 208?
- ⑦ Find the number which, when added to its square, makes 90.
- ⑧ Twice the square of a certain whole number added to 3 times the number makes 90. Find the number.
- ⑨ The area of a rectangle is 60 cm^2 . The length is 11 cm more than the width. Find the width.
- ⑩ A rectangular plot measures 12 m by 5 m. A path of constant width runs along one side and one end. If the total area of the plot and the path is 120 m^2 , find the width of the path. (Hint: Make a sketch and let the width of the path be $x \text{ m}$.)
- ⑪ A rectangular piece of cardboard measures 17 cm by 14 cm. Strips of equal width are cut off one side and one end. The area of the remaining piece is 108 cm^2 . Find the width of the strips removed.
- ⑫ A man is 37 years old and his child's age is 8. How many years ago was the product of their ages 96?